

$$= (2.0 \times 10^9)(\pi)(10)(0.30^2 - 0.15^2)(9.81)(5.0) = 208 \text{ kJ} \sim 210 \text{ kJ}$$

7 Answer: B

$$F_c = W - N$$

$$\text{For } N = 0$$

$$mv^2/r = mg \rightarrow v^2 = (23)(9.81) \rightarrow v = 15$$

8 Answer: B